

AI SDLC Studio

Autonomous Multi-Agent Platform for the Software Development Lifecycle

Full Project Overview

Functional & Architectural Summary — Implementation Details Withheld

1. Executive Summary

AI SDLC Studio is a full-stack application that automates the early-to-mid stages of building software. A user describes a business problem in plain language, and the system guides that idea through four stages — Requirements, Design, Development, and Testing — producing professional-grade documentation and a working application at the end of the process.

Each stage is powered by a coordinated set of AI agents and is followed by a mandatory human review checkpoint, so a person always approves or sends back the output before the project advances. This document describes what the platform does and how it is put together at a conceptual level. Internal engineering details — such as source code, prompt design, and service-level implementation — are intentionally not included.

2. Project Objective

The platform is built to close the gap between a business idea and a working piece of software, which traditionally requires a business analyst, a solution architect, and a development team working over days or weeks. AI SDLC Studio compresses this into a single guided workflow that a business user can drive directly, while still keeping a human decision-maker in control at every stage.

Core Goals

- Convert an unstructured problem statement into a structured, professional requirements specification.
- Identify gaps or ambiguities in requirements automatically and resolve them through guided clarification.
- Produce a complete system design, including architecture, data design, interface design, and visual diagrams.
- Generate a working full-stack application aligned to the approved design.
- Generate and run a layered set of automated tests, with a downloadable results report.
- Maintain a complete, versioned audit trail of every stage, decision, and reviewer comment.
- Allow every artifact — requirements, design, code, and tests — to be exported in common formats.

3. High-Level Architecture

The platform follows a client–server model, made up of two independent applications that communicate over a secured connection.

- **Client Application:** A web-based interface where users hold a chat-style conversation with the system, review generated artifacts, view diagrams, and approve or reject work at each stage.
- **Server Application:** A backend service that coordinates the AI agents, stores project data, and generates downloadable documents and reports.
- **AI Reasoning Layer:** A pluggable connection to a large language model provider that powers the generation of requirements, designs, code, and tests. The platform is designed to keep functioning even when this layer is temporarily unavailable.

- **Data Store:** A structured database that retains every project, version, review decision, and activity log for full traceability.

The internal orchestration logic, prompt design, and code-level implementation that power these components are considered proprietary and are not detailed in this document.

4. The Four-Phase Workflow

Every project moves through four phases. Each phase produces a clear deliverable and ends with a human approval checkpoint before the next phase can begin.

Phase	What It Delivers
1. Requirements	A structured specification of business goals, users, workflows, and functional and non-functional requirements, refined through guided clarification and a finalized backlog of epics, features, and user stories.
2. Design	A complete system design covering architecture approach, technology direction, data design, interface design, security approach, and a set of visual diagrams illustrating how the solution fits together.
3. Development	A working full-stack application generated in line with the approved design, covering the interface, business logic, and data layers, along with supporting documentation.
4. Testing	A layered set of automated tests covering multiple types of checks, executed against the generated application, with a summarized results report available for download.

At the end of each phase, a reviewer can approve the output to proceed, or reject it with comments — in which case the relevant phase is refined and re-presented for review, rather than requiring the whole project to start over.

5. Human Oversight & Review Gates

A defining principle of the platform is that no phase is allowed to complete without a human decision. This is enforced through structured review checkpoints at each meaningful step of every phase — for example, after requirements are extracted, after gaps are identified, after a quality check is performed, and after a design or build is generated.

- Reviewers can approve a stage to move the project forward.
- Reviewers can reject a stage and provide comments, which are used to refine the next attempt.
- For the generated application, reviewers can approve or request changes to specific parts of the build, rather than the entire result, allowing for targeted, efficient revisions.
- Every review decision, along with reviewer comments and timestamps, is permanently recorded.

6. Key Platform Features

Feature Area	Description
Guided Requirements Chat	A conversational interface that captures a business idea and turns it into a structured specification, including the ability to upload existing documents as a starting point.
Automatic Gap Detection	Identifies missing details, ambiguities, or contradictions in requirements and raises clarifying questions before finalizing.
Quality Scoring	Requirements are checked for completeness and clarity before being allowed to proceed to design.
Visual System Design	Produces an easy-to-read set of diagrams illustrating system structure, data relationships, user flows, and deployment layout.
Full Application Generation	Produces a working application spanning interface, logic, and data layers, aligned to the approved design.
Targeted Revisions	Allows specific parts of a generated build to be revised without regenerating the whole application.
Automated Test Suite	Generates and runs a layered set of tests, with categorized pass/fail results and a downloadable report.
Full Traceability	Every requirement can be traced through design and into the final build, supporting confidence and audit needs.
Version History	Every phase retains a complete history of prior versions and reviewer feedback.
Flexible Exports	Requirements, design, and test documentation can be exported in common formats such as PDF, Word, and Markdown.
Resilience	The platform is designed to remain usable and demoable even during temporary interruptions to the underlying AI service.

7. Security & Data Handling (General Overview)

Access to the platform's backend services is protected through a controlled authentication mechanism, and all requests are restricted to approved application origins. Every project's data — including requirement versions, design versions, generated builds, and review history — is stored in a structured, versioned data store, giving a complete and auditable record of how each project evolved.

Incoming user input also passes through a set of protective checks before being processed, helping ensure the platform is used for its intended purpose and reducing the risk of misuse. The specific mechanisms behind these protections are internal implementation details and are not covered in this document.

8. Business Benefits

- Significantly faster turnaround from idea to working software.
- Reduced dependency on the simultaneous availability of multiple specialists.
- Higher confidence through mandatory review gates and full traceability.
- Clear, exportable documentation suitable for stakeholders at any technical level.
- A repeatable, consistent process that scales across multiple projects.

9. Current Limitations

- Every phase still requires human review — this is by design, not a fully unattended process.
- Best suited to well-defined, small-to-mid-sized solutions rather than very large enterprise systems.
- Currently designed for a single team or organization rather than many isolated customer accounts.
- Generated tests provide strong initial coverage and are intended as a solid starting point, not a substitute for full independent verification before production use.

10. Future Roadmap

- Support for multiple, separately secured customer accounts.
- Deeper integration with commonly used business and project-management tools.
- Support for larger and more complex solution builds.
- Expanded reporting and dashboards across multiple projects.
- Additional automated quality and reliability checks on generated builds.

11. Summary

AI SDLC Studio guides a business idea through requirements, design, development, and testing, using coordinated AI agents at each stage and a mandatory human approval gate before any stage is considered complete. The result is a faster, more transparent, and fully traceable path from concept to working software — with people, not just automation, in control of the outcome at every step.

This document intentionally excludes internal engineering details such as source code, internal service design, and AI prompt structures, which are considered proprietary to the project.